

Course report
SF2321 Applied Statistical Analysis
Fall 2023
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Introduction

The course SF2321 Applied Statistical Analysis is a second-cycle method course for MA students in Criminology¹, European Studies (MAES), International Administration and Global Governance (IAGG), Political Communication (POLCOM), Political Science (MAPS), Political Science: Environmental Governance and Behavior (MAPSE) and Sociology.² The 2023 course edition has undergone significant revisions in both learning outcomes and content, thus 2023 was the first time the course was held in its current form.

Content-wise, the course has shifted focus from theory-based data analysis to research design and statistical techniques for causal inference. Two-thirds of the learning outcomes in the syllabus have been revised. Compared to the course version held 2018-2022, content connected to 4 hec is completely new and content for the remaining 11 hec is significantly updated. In short, causal inference design and the potential outcomes framework is a red thread throughout the entire course. Technique-wise, we removed categorical regression and multi-level regression in favor of the difference-in-difference method and the synthetic control method. The grading criteria for the method thesis were also revised.

Four teachers, all associate professors, from the Department of political science taught on the course: Mattias Agerberg, Rasmus Broms, Maria Solevid and Aksel Sundström. As usual, all teachers were involved in supervision and examination of the method theses.

At the start of the course 2023, 32 students were registered on the course, 31 first-time registered and 1 re-registered³. All of the first-time registered were active and took credits. Of the 31 first-time registered and active students, 3 were from IAGG, 12 from MAPS, 7 from MAPSE, 2 from POLCOM, 5 from Sociology and 2 free-standing students.

The course was taught on campus with the addition of some well-thought through digital teaching events. Based on previous student cohorts positive feedback, both during and after the pandemic, we have deliberately kept the pre-recorded introduction to the Stata tutorials, where the instructor demonstrates the code and procedures in Stata relevant for the assignment (1-6). We also carry out the two advice sessions related to the Method thesis on Zoom, this as these sessions take place just before, and in the middle of, the Christmas and New Year holiday.

¹ MA students in Criminology participate in one of the two quantitative method courses every other year (2019, 2021 and so on).

² The course started in 2007 and was then a formal collaboration between Department of Journalism, Media and Communication, Department of Sociology and Work Science and Department of Political Science (coordinating department), but the participating MA programs has shifted over the years. Throughout the years, the course has been revised several times, and big revisions took place in 2012 and 2018.

³ Two additional students re-registered during the course. Also, three students who started the course changed to SF2324 after two weeks.

Course structure

Table 1 below summarizes the course structure of the 2023 course which had a new structure compared to the 2018-2022 course editions.

Table 1 Overview of course structure, 2023

	Course structure	Responsible teacher
Week 1-2	Introduction Assignment 1: Design for causal inference (1 hec)	Group assignment, oral presentation at compulsory seminar Maria Solevid/Mattias Agerberg
Week 2-3	Recap and Introduction to Stata	Aksel Sundström
Week 3-4	Assignment 2: Introduction to linear regression (2,5 hec)	Group assignment, paper Aksel Sundström
Week 4+5	Assignment 3: Heterogenous effects in linear regression (1,5 hec)	Group assignment, paper Maria Solevid
Week 5	Assignment 4: Panel regression (1 hec)	Group assignment, paper Rasmus Broms
Week 6	Assignment 5: Difference-in-difference (1,5 hec)	Group assignment, paper Rasmus Broms
Week 7	Assignment 6: Synthetic controls (1,5 hec)	Group assignment, paper Mattias Agerberg
Week 8-10	Method thesis (6 hec)	Individual assignment, paper and oral presentation at compulsory seminar Maria Solevid (all teachers involved)

The course starts with a course introduction and a recap lecture on statistical inference, followed by lectures related to assignment 1: designs for causal inference. This smaller assignment is examined at a mandatory seminar where the students shortly presents a research design. The lectures and assignment aim introducing and training students how to set up causal research questions, how to formalize the expectations and the counterfactuals using the potential outcomes framework. The potential outcomes framework returns in all lectures and assignments throughout the course.

After the first assignment, we provide a recap lecture on the variable language, the data matrix, variable/variable units, descriptive statistics, cross-tabs and correlations as well as tutorial that introduces the software Stata. Once this is finished, we move on to linear regression (bivariate and multiple regression) and the different parameters estimated, dummy variables and a shorter introduction to regression diagnostics. We also introduce the Unexpected event during survey design (UESD), a quasi-experimental design using cross-sectional survey data to reach causal conclusions. We continue with linear regression during the workshop on heterogenous effects in linear regression in which we teach how to think, use, interpret and understand regression models with an interaction term. UESD comes back side by side with many other examples.

After these $\approx 2,5$ weeks focusing on linear regression, the course moves on to applied methods for causal inference: panel regression, difference-in-difference and synthetic controls. The panel regression workshop introduces how to handle repeated observations over time and the difference between pooled OLS, fixed effects and first-difference models. The difference-in-difference workshop introduces the design for how to compare trends in two groups (countries, municipalities, individuals) when one of the groups is exposed to a treatment, often a natural experiment such as a reform, and how to use panel regression (and cross-sectional data) to test this empirically. Finally, the synthetic control method introduces how

one can design and test the causal effect of a specific event on a single unit (such as the German unification, or the effect of a new tax in one US state) by creating and comparing with matching cases that were not treated (for example, other European countries, or other US states).

As earlier, the course ends with the individual assignment, the method thesis.

Completion rate

The completion rate at the end of the course, including the revision round for those that didn't reach pass, but also didn't receive fail, was 81 percent among the first-time registered students (78 percent among all registered students). The completion rate among all students is higher compared to the previous course version held 2018-2022. During these years, completion rates varied between 53 and 69 percent.

Of the 25 students who had completed the course during the original submission rounds, eight were awarded Pass with distinction, 16 Pass and one Fail. The share of Pass with distinction is 32 percent which is higher than 2022 and 2021 (23 percent and 22 percent respectively) and on par or lower compared to 2018-2020 (2020: 31%, 2019: 42%, 2018: 38%).

Students' assessment of core course elements

Of the 32 registered students, 21 answered the course evaluation survey which equals a response rate of 65,6 percent. The response rate is the second highest since 2018.⁴ If we consider that 31 students have been active throughout the course, a more accurate response rate is 67,7 percent. Time to fill out the course evaluation scheduled at the end of the final seminars, and students were also verbally reminded at the end of the seminar to answer the survey. But since the seminars went on from 13:00 to 17:00, fewer students stayed to fill out the survey (we don't blame them, we were tired too!). Invitation and reminders to the survey were as usual sent out through the SUNET survey system, and one additional reminder was made by the course coordinator through Canvas messages.

Table 2 summarizes the students' assessment of the course in general, the lectures, teachers, computer labs and overall course organization/administration. Most tables present both percentages and means and were applicable, we also include the aggregated average from 2018–2022 for comparison. Please keep in mind that the 2023 course significantly differs from the 2018-2022 course.

⁴ RR 2022: 53,7 percent of all/59,5 percent among active (course evaluation scheduled in seminar program)
RR 2021: 46 percent of all/61 percent among active (verbal reminder during Zoom seminar to fill out the course evaluation survey after the seminar)
RR 2020: 53 percent among all/76 percent among active (verbal reminder during Zoom seminar to fill out the course evaluation survey after the seminar)
RR 2019: 70 percent among all (scheduled time to fill out course evaluation in classroom)
RR 2018: 59 percent among all (verbal reminder at the end of the seminar to fill out the course evaluation survey)

Table 2 General assessments of the course, lectures, tutorials, teachers and course administration (percent and mean)

	The course in general	Lectures	Stata tutorials	Teachers	Organization and administration of the course
Very good	67	38	29	57	62
Rather good	24	57	52	38	24
Neither bad nor good	9	5	9,5	5	9
Rather bad	0	0	9,5	0	5
Very bad	0	0	0	0	0
Total	100	100	100	100	100
N	21	21	21	21	21
Mean 2023	4,6	4,3	4,0	4,5	4,4
Mean 2018-2022	4,2	3,9	4,2	4,3	4,5

Comment: Question formulations: "Overall, how would you assess the....?". Very bad coded as 1 and Very good coded as 5. For the question on distance teaching, the question wording is "In general, how do you think the distance teaching worked?".

Overall, the core course elements receive positive assessments. 67 percent responds that the course was very good and 24 percent answers rather good. In total, this means that 91 percent provides a positive assessment (very or rather good). The mean on a 1-5 scale, were 1 equal very bad and 5 equal very good, is 4,6 which in fact is the highest average assessment in comparison to the previous course versions between 2018-2022(both higher than the aggregated mean and higher than each individual course edition). 95 percent respond that the lectures are rather or very good (mean 4,3), 81 percent shares the same view of the Stata tutorials (mean 4,0) and 95 percent make positive assessments of the teachers (mean 4,5). 86 percent assesses the organization and administration of the course as rather or very good (mean 4,4). Overall, the 2023 average assessments are either higher or similar to the aggregated means from the previous years.

Students' assessment of the different workshops

Table 3 summarizes students' assessments of the different workshops throughout the course. Since all workshops are revised or new, we don't make any comparisons to the previous course versions.

Table 3 Evaluations of the different workshops (percent and mean)

	Very good	Rather good	Neither good nor bad	Rather bad	Very bad	Total	N	Mean
Design for causal inference (A1)	65	30	5	0	0	100	20	4,6
Recap and introduction to Stata	35	35	15	15	0	100	20	3,9
Introduction to linear regression (A2)	10	55	15	20	0	100	20	3,5
Heterogenous effects in linear regression (A3)	70	15	15	0	0	100	20	4,6
Panel regression (A4)	35	40	20	5	0	100	20	4,0
Difference-in-difference (A5)	40	45	10	5	0	100	20	4,2
Synthetic controls (A6)	55	25	20	0	0	100	20	4,4
Method thesis (A7)	43	33	19	5	0	100	21	4,1

Comment: Question formulations: "What is your assessment of the different workshops specified below?" Very bad coded as 1 and Very good coded as 5. Recap and introduction to Stata is the only workshop not associated with an assignment.

The general conclusion is that all workshops receive positive evaluations with average assessment ranging from 3,5 to 4,6. The workshops on *Recap and Stata intro* (no assignment) and *Introduction to regression* respectively stands out a little, being the only ones with an average below 4. However, both these workshops show clearly more positive than negative assessments.

If we focus on the workshops connected to assignment 1-6, the shares of students responding that the workshop was very good varies from 10 to 70 percent, with the lowest score for *Introduction to linear regression* and the highest scores for *Design for causal inference* and *Heterogenous effects in linear regression* respectively. If we combine the Very good and rather good response options, the share of positive assessments are 95 percent (Design for causal inference), 65 percent (Introduction to linear regression), 85 percent (Heterogenous effects), 75 percent (Panel), 85 percent (Diff-in-diff), 80 percent (Synthetic control) and finally 76 percent (Method thesis). We are especially happy to see that all the brand-new workshops (Design for causal inference, Diff-in-diff and Synthetic control) receive positive assessments, with averages between 4,2 and 4,4.

Students' assessments of examination forms, literature, and amount of work

As shown in table 4, the examination forms – the six workshop assignments and the method thesis– receives average assessments of 4,4 and 4,3 respectively. 90 and 85 percent of the students respond that these examination forms are rather good or very good. The workshop assignments are assessed slightly more positive in 2023 compared to five-year average 2018-2022 while the assessment of the method thesis as examination form is on par with the five-year average.

Table 4 Evaluation of examination forms

	Workshops assignments as examination forms	Method thesis as examination form
Very good	52	52
Rather good	38	33
Neither bad nor good	10	10
Rather bad	0	5
Very bad	0	0
Total	100	100
N	21	21
Mean 2023	4,4	4,3
Mean 2018-2022	4,1	4,2

Comment: Question formulation: Overall, how would you assess the examinations? Very bad coded as 1 and Very good coded as 5

The main course literature consists of two textbooks: *Mastering Metrics* by Joshua Angrist and Jörn-Steffen Pischke (new book) and *Applied statistics using Stata* by Mehmet Mehmetoglu and Tor Jakobsen. In short, the books receive weakly positive lukewarm assessments (see table 5). The book by Mehmetoglu and Jakobsen is notably much less popular compared to previous years which probably can be related to that this book does not “speak” the causal inference language and that fewer chapter are relevant when the course mainly uses OLS based regression techniques. The free-text responses indicates that the students feel neither of the books were perfect, and that students feel they didn’t need/have time to read the literature (this also includes the various research articles on the literature list).

Table 5 Evaluation of the literature

	Angrist & Pischke Mastering Metrics	Mehmetoglu & Jakobsen Applied statistics using Stata
Very good	24	14
Rather good	33	38
Neither bad nor good	38	33
Rather bad	5	10
Very bad	0	5
Total	100	100
N	21	21
Mean	3,8	3,5
Mean 2018-2022		4,4

Comment: Question formulation: "Overall, how would you assess the literature?" followed by the two author names and titles of the two course books. Very bad coded as 1, Very good coded as 5. In addition to the response options displayed in the table, the option "I didn't read this book" was also available but has been excluded from the mean calculations.

The assessment regarding amount of work and how many hours the student spent on course work (inside and outside the classroom) is reported in table 6. The results most notably show that 85 percent think it the course work was somewhat too demanding or too demanding and only 14 percent says the workload was about right. 38 percent responded they spent more than 40 hours per week on the course work, 43 percent answers 31-40 hours and 19 percent 21-30 hour. Most of the students have, however, not spent more than 40 hours on course work. Judging from the displayed averages, the course was viewed as more demanding compared to previous years, and students also reports spending a lot of time on course work. This is also notable in the free-text answers.

Table 6 Evaluation of amount of work

Amount of work		How much time spent on course work	
Too demanding	33	More than 40 hours	38
Somewhat too demanding	52	31-40 hours	43
About right	14	21-30 hours	19
Not that demanding	0	11-20 hours	0
Not at all demanding	0	Less than 10 hours	0
Total	100	Total	100
N	21	N	21
Mean 2023	4,2	Mean 2023	4,2
Mean 2018-2022	3,7	Mean 2018-2022	3,9

Comment: "What is your assessment of the amount of work you have been requested to carry out during the course?". Not at all demanding coded as 1, Too demanding coded as 5 "On average, approximately how much time did you spend on coursework, both inside and outside the classroom, per week?". Less than 10 hours coded as 1, More than 40 hours coded as 5.

Students' learning and expectations

67 percent of the students answer that they experienced a Eureka moment during the course (not shown in table). Among the relatively few free-text answers, the potential outcomes framework and the importance/usage of causal study designs are the more common answers.

In the discussion and research about course evaluations, there is a drive towards a transition from pure product assessments (how satisfied the students are with the 'delivery' of the course or the 'performance' of teacher) to assessments based on students' own learning process. This implies a shift from quality control to quality development. As stated by

Zerihun et al (2012:102), "students will be in a better position to provide appropriate feedback for teaching improvement if the evaluation of teaching quality also addresses their learning experiences".⁵

To facilitate student reflections about their own learning, I initiated beehive discussions at the course introduction lecture. The students were asked to in small groups discuss two questions: 1) What do you expect (or want/wish) to learn during the course? 2) How do you plan to achieve the expectations regarding your own learning? and to write down their answers using Menti.

To follow-up this introductory group discussion, I included a text in the course evaluation with the intent to remind students about these initial discussions, followed by a question of expectation fulfilment. 57 percent of the students responded that their expectations were fulfilled, 24 percent that expectations were more than fulfilled and 19 percent that expectations were partially fulfilled (see table 7). The average assessment of expectation fulfilment is on par with previous course versions.

Table 7 **If you think back to your initial reflections and discussion about your own learning expectations of the course, to what extent would you say your expectations have been fulfilled? (percent and mean)**

Expectations more than fulfilled	24
Expectations fulfilled	57
Expectations partially fulfilled	19
Expectations not fulfilled	0
Total	100
N	21
Mean 2023	3,1
Mean 2019*-2023	3,0

Comment: Question reads: "At the course introduction, you were asked to in small groups reflect and discuss your learning expectations and how you, through your own learning, should achieve those expectations. If you think back to your initial reflections and discussion about your own learning expectations of the course, to what extent would you say your expectations have been fulfilled?" In addition to the response options displayed in the table, option "No opinion/I didn't attend the course introduction" was also offered but no students answered this. Expectations not at all fulfilled coded as 1, Expectations more than fulfilled coded as 4. *The 2019 average is the combined average from the two experiment groups

Reflections

From our teacher perspective, we are overall very satisfied with how the course went. Considering how much content that was revised or new, we are very happy to see that the course also was well-received among the students. We think that one important key why the course was successful is that we in the team of teachers spent a lot of time discussing the course content, everything from learning outcomes down to the sequence of the different workshops and assignments. We also put considerable effort into aligning the different lecture and assignment contents with each other. The process started already during the spring of 2023 when we outlined the new course structure in detail and revised the syllabus. We then worked with the detailed content in October and November.

However, there are obviously still room for improvement. From the free-text answers in the course evaluations, students mention that the pace during the course is uneven, that it is

⁵ Zerihun, Z, Beishuizen, J & Van Os, W. 2012. Student learning experience as indicator of teaching quality. *Educational Assessment, Evaluation and Accountability*, 24:99-111. Please also note that this text paragraph appears in identical form in previous course reports.

generally too much work, that Stata tutorials could have been better organized and that we should provide more examples of the potential outcome framework throughout the course.

Taken the students' responses and our own thoughts into account, we are currently discussing several improvements. Among other things, we are thinking about introducing a diagnostic test so that the students can self-assess their previous knowledge about statistical methods to make sure students have the necessary knowledge we expect that they should have, we are considering re-making the recap lecture into several shorter videos that students are prompted to watch and combined this with a Q&A session afterwards (flipped classroom), we will better introduce the different methods in the beginning of the course, we are discussing whether the first memo for the method thesis should even be a bit later (despite student responses on that we should start earlier) and to replace the memo (with written feedback) and the q&a session two weeks later, with one session where students almost fill out a form about their topic and then shortly meets with supervisor. The motivation behind this change is that we want more students to conduct method theses with a proper causal inference design, some of which are introduced quite late in the course. We will also make revisions in the content related to introduction to regression and make sure we provide more examples of notation and thinking behind the potential outcomes framework. We will also make sure students have slightly more time for the more advanced methods, in particular panel regression. We will also try to have as many submissions as possible on Fridays to make sure students get more free-time.